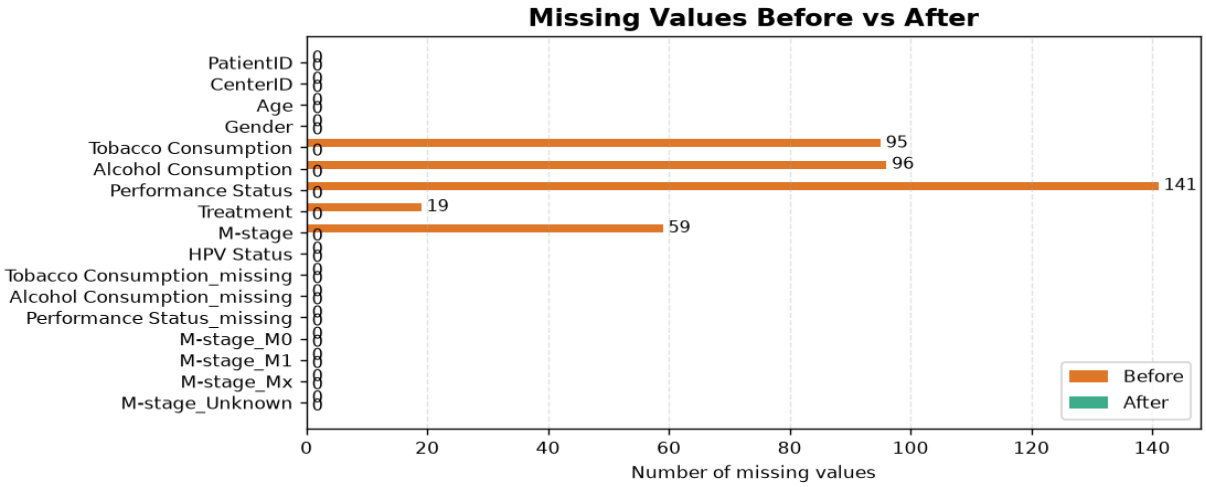
PREPROCESSING ENTRY DATA

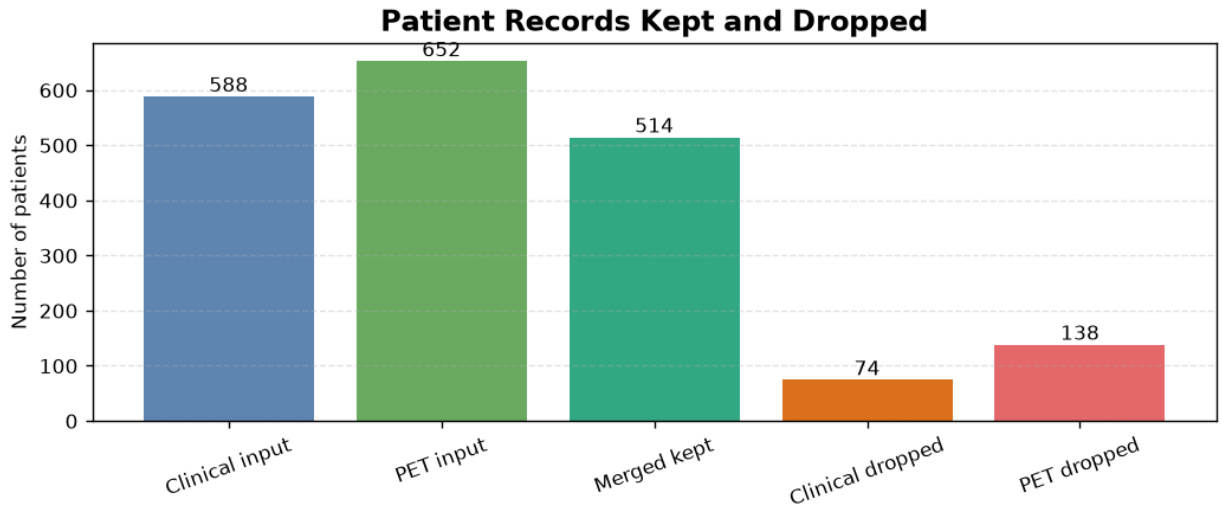
Methods used :

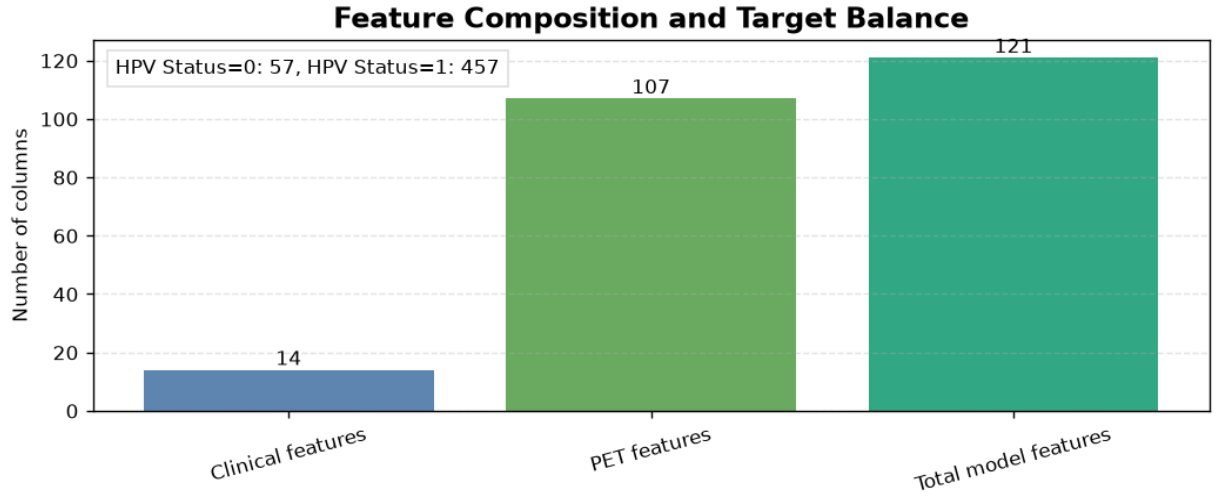
1. **Rule-Based Data Cleaning**
   Invalid values in **Performance Status** are converted to missing values (NaN), while only valid categories (0–4) are retained. This ensures data consistency before further preprocessing.
2. **Missing Indicator Method**
   Binary indicator columns are created to record whether values were originally missing before imputation. This preserves potentially useful information about missing data patterns.
3. **Median Imputation**
   Missing values in the **Age** feature are replaced with the median calculated from the training set. The median is robust to outliers and suitable for continuous numerical variables.
4. **Most-Frequent Imputation**
   Missing categorical and discrete clinical variables are filled with the most common value (mode) from the training set. This maintains valid category values while minimizing bias.
5. **Constant Imputation**
   Missing values in **M-stage** are replaced with the fixed category **"Unknown"**. This treats missingness as a separate meaningful category rather than removing information.
6. **Min–Max Normalization**
   After imputation, the **Age** feature is scaled using Min–Max normalization based on the training set minimum and maximum values. This transforms Age into a fixed range (typically 0–1) while preserving the relative differences between observations.
7. **One-Hot Encoding**
   The categorical **M-stage** variable is converted into separate binary columns for each category, including **Unknown**. This allows machine learning models to process categorical data without assuming any ordinal relationship.



Merging clinical + pet :

- **Inputs used**
  Clinical: clinical_entry_preprocessed.csv
  PET: pet_features.csv
- **Merge key**
  Both files were matched by **PatientID**.
- **Join type**
  Inner join was used. This keeps only patients present in both files.
- **Dropped records**
  Clinical rows without PET features: **74**
  PET rows without clinical records: **138**
- **MLP readiness check**
  Missing cells in merged CSV: **0**
  Non-numeric feature columns: **none**
  Model features if **PatientID** is removed and **HPV Status** is the label: **121**





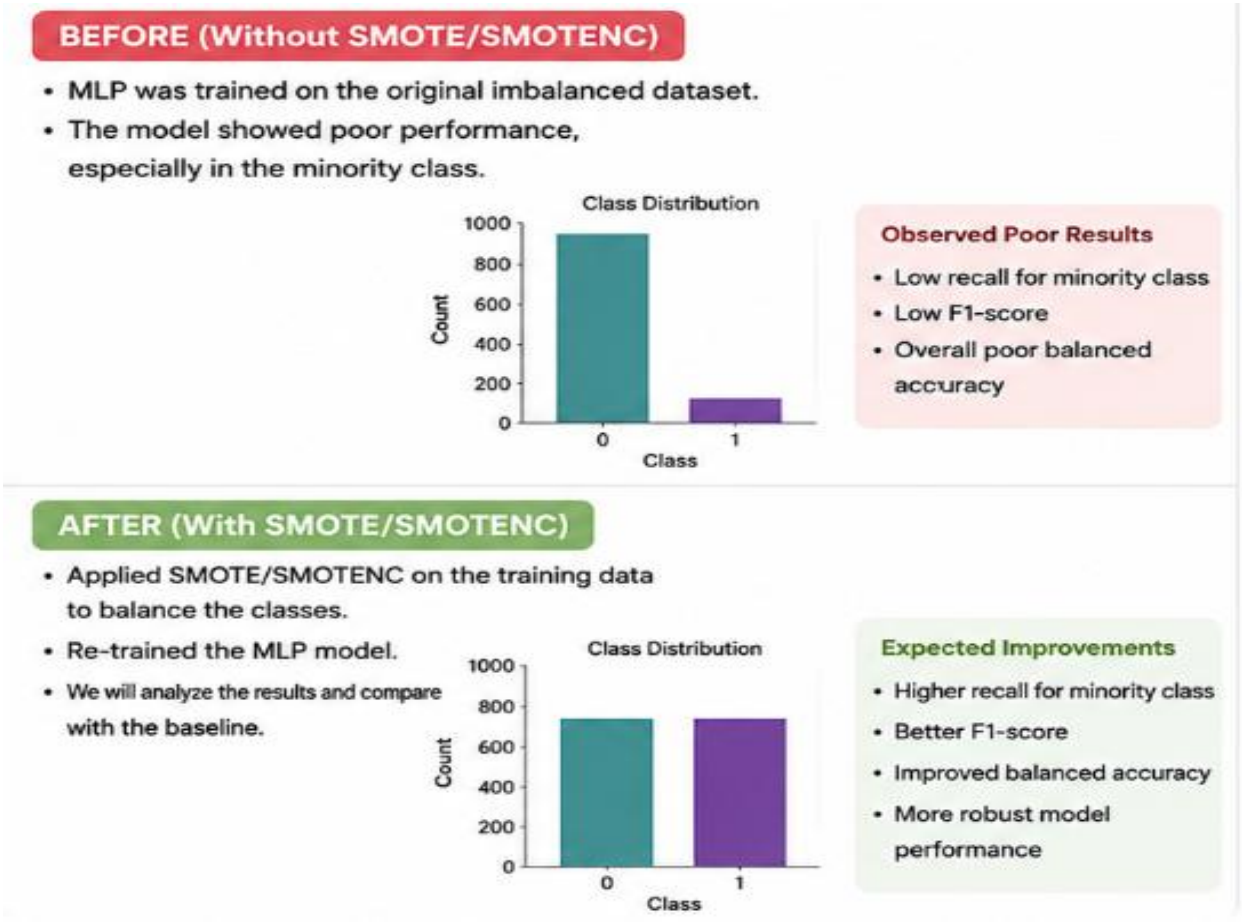
MLP BINARY CLASSIFICATION :

SMOTE and SMOTENC for MLP Classification

In the initial MLP classification experiments, the model was trained using the original imbalanced dataset without applying any oversampling techniques. The obtained results showed relatively poor performance, particularly in detecting the minority class, indicating that class imbalance negatively affected the model's learning ability.

To address this issue, two oversampling methods were introduced: **Synthetic Minority Oversampling Technique (SMOTE)** and **SMOTENC (SMOTE for Nominal and Continuous features)**. SMOTE generates synthetic samples for the minority class by interpolating between existing minority instances, thereby balancing the class distribution. SMOTENC extends this approach by supporting datasets that contain both continuous and categorical features, making it more suitable for mixed clinical data.

The MLP classification was then repeated using these oversampling techniques on the training data. The resulting performance will be analyzed and compared with the baseline model to evaluate the impact of SMOTE and SMOTENC on classification accuracy, minority class detection, and overall model performance.



CLINICAL ONLY :

Metric	Validation	Test
ROC-AUC	0.978	0.862
Accuracy	0.922	0.872
Balanced Accuracy	0.957	0.783
Sensitivity	0.913	0.899
Specificity	1.000	0.667
Precision	1.000	0.954
F1-score	0.955	0.925
Confusion Matrix	[[8, 0], [6, 63]]	[[6, 3], [7, 62]]
Brier Score	0.048	0.089
PR-AUC	0.997	0.969
Expected Calibration Error (ECE)	0.058	0.106

1. The clinical-only model is the strongest individual model. The validation ROC-AUC is 0.978 and the test ROC-AUC is 0.862, showing that the clinical features contain strong information for predicting HPV Status. The decrease from validation to test means the model generalizes well, but not perfectly.

2. Accuracy is high on both splits. Validation accuracy is 0.922, meaning 71 of 77 validation patients were classified correctly. Test accuracy is 0.872, meaning 68 of 78 test patients were classified correctly. This shows strong overall classification performance.

3. Balanced Accuracy is also strong, especially on validation. Validation Balanced Accuracy is 0.957, which means the model performed well for both HPV-negative and HPV-positive classes. Test Balanced Accuracy is 0.783, which is lower than validation but still useful. The drop is mainly caused by weaker test specificity.

4. The model keeps high sensitivity on both validation and test. Validation sensitivity is 0.913, and test sensitivity is 0.899. On the test set, this means the model correctly detected 62 of 69 HPV-positive patients and missed 7 HPV-positive patients.

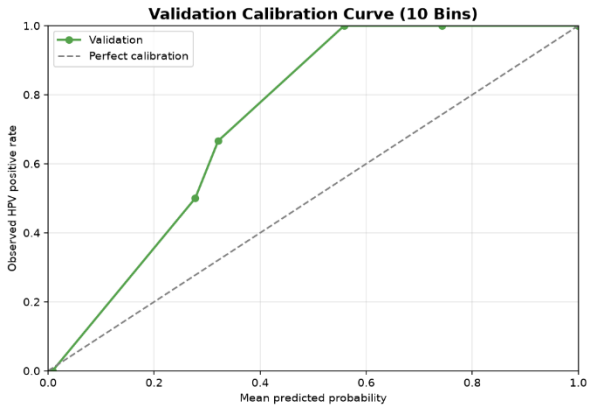
5. Specificity is excellent on validation but lower on test. Validation specificity is 1.000, meaning all 8 HPV-negative validation patients were correctly identified. Test specificity is 0.667, meaning 6 of 9 HPV-negative test patients were correctly identified, while 3 were false positives.

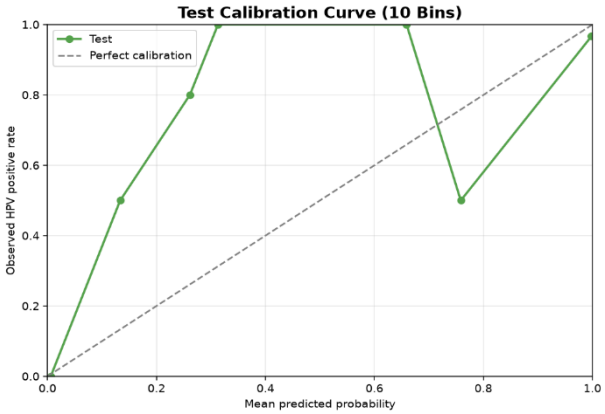
Validation Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	4	0.008	0.000	0.008
[0.1, 0.2)	0			
[0.2, 0.3)	4	0.277	0.500	0.223
[0.3, 0.4)	6	0.321	0.667	0.346
[0.4, 0.5)	0			
[0.5, 0.6)	1	0.558	1.000	0.442
[0.6, 0.7)	0			
[0.7, 0.8)	4	0.743	1.000	0.257
[0.8, 0.9)	0			
[0.9, 1.0]	58	1.000	1.000	0.000

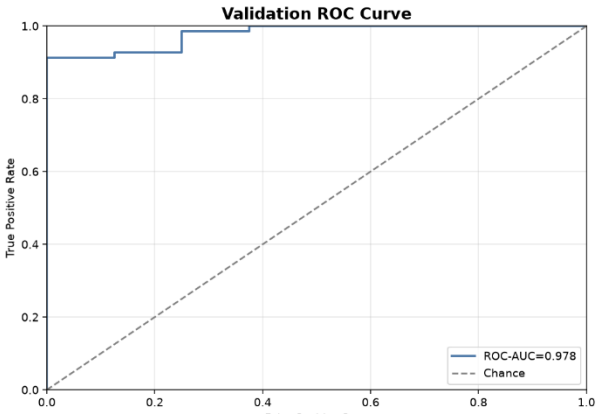
Test Calibration Table (10 Bins)

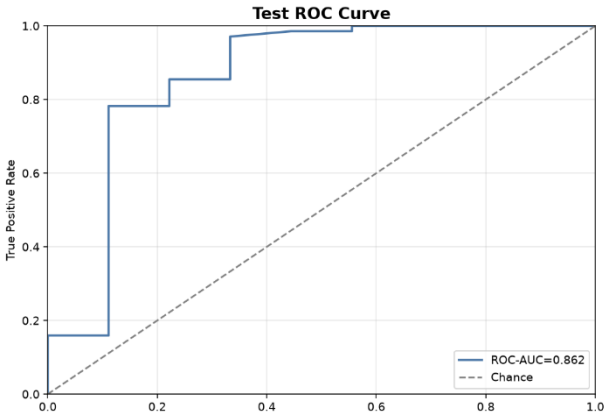
Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	4	0.006	0.000	0.006
[0.1, 0.2)	2	0.133	0.500	0.367
[0.2, 0.3)	5	0.261	0.800	0.539
[0.3, 0.4)	2	0.313	1.000	0.687
[0.4, 0.5)	0			
[0.5, 0.6)	0			
[0.6, 0.7)	3	0.659	1.000	0.341
[0.7, 0.8)	2	0.758	0.500	0.258
[0.8, 0.9)	0			
[0.9, 1.0]	60	0.998	0.967	0.032



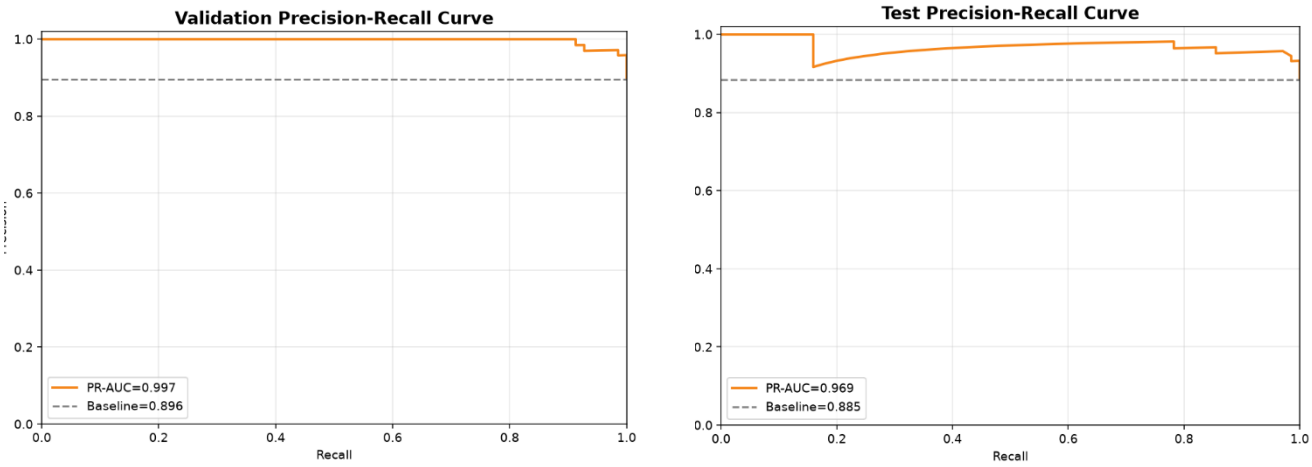


The validation ECE is 0.058 and the test ECE is 0.106. The test calibration is weaker than validation, but still acceptable compared with PET-only before scaling. The Brier Score is 0.048 on validation and 0.089 on test, which supports the same idea: the clinical probabilities are fairly useful, but the test probabilities are less calibrated than validation.

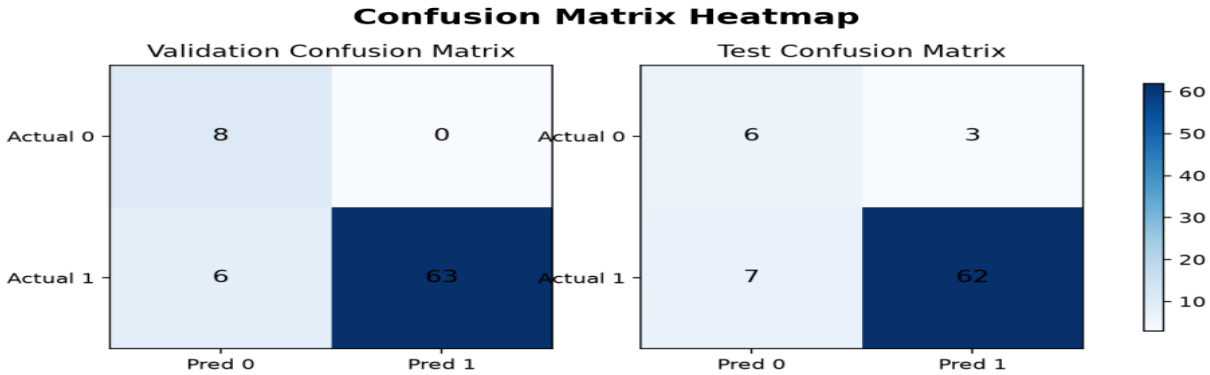




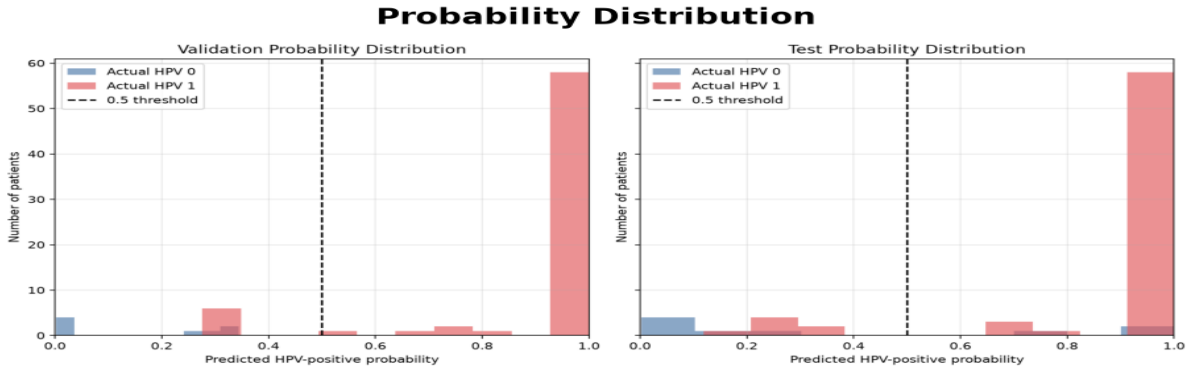
The ROC curves show strong discrimination. Validation ROC-AUC is 0.978, which is very high. Test ROC-AUC is lower at 0.862, but still good. This means the clinical model usually ranks HPV-positive patients above HPV-negative patients, although the performance drops on unseen test data.



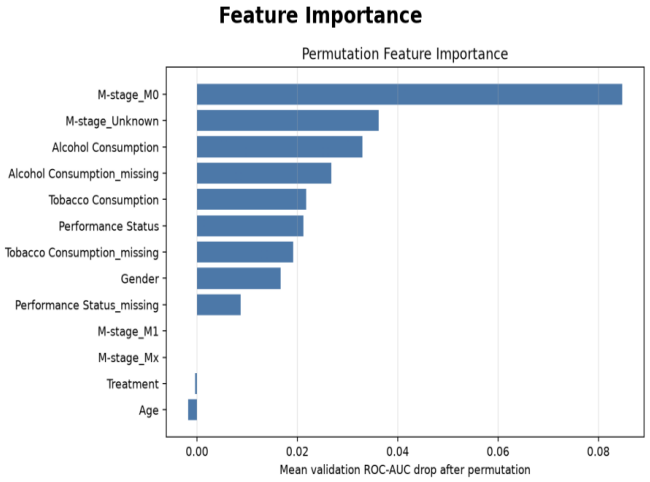
The PR-AUC is 0.997 on validation and 0.969 on test. Because the dataset has many more HPV-positive patients than HPV-negative patients, the precision-recall curve is important. These high PR-AUC values show that the clinical model keeps strong precision while recovering most HPV-positive patients.

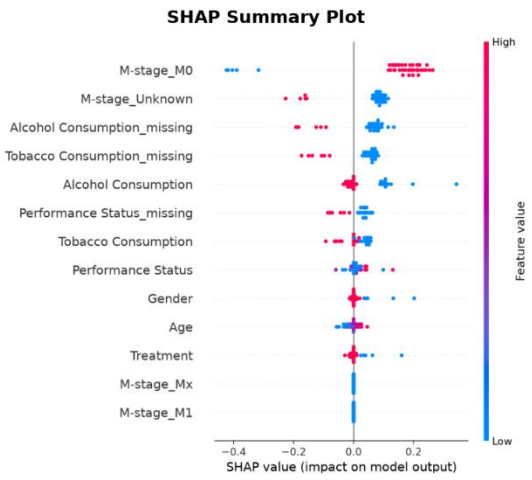


On validation, the model correctly identified all 8 HPV-negative patients and 63 of 69 HPV-positive patients. It made 0 false positives and 6 false negatives. On test, it correctly identified 6 of 9 HPV-negative patients and 62 of 69 HPV-positive patients. It made 3 false positives and 7 false negatives.



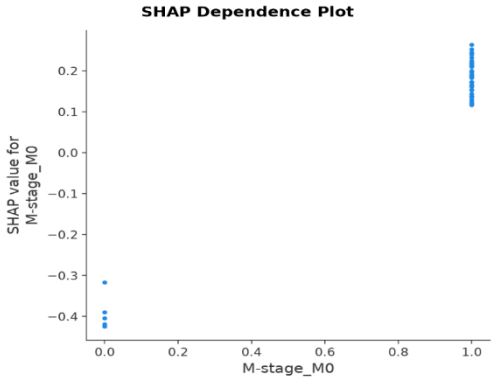
The probability distribution plot shows how predicted HPV-positive probabilities are separated by true class. For the clinical model, HPV-positive patients are generally pushed toward high predicted probabilities. HPV-negative patients are more often lower, but there is some overlap on test, which explains why test specificity is 0.667 rather than perfect.

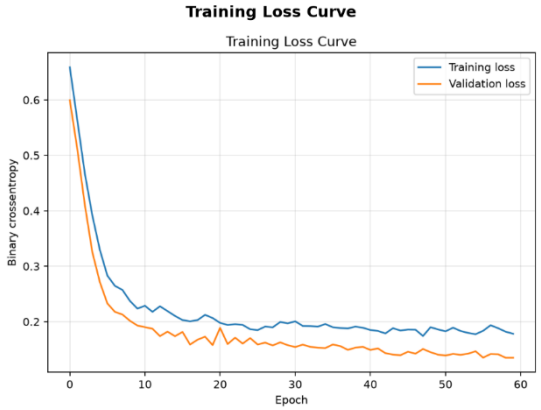




The permutation feature importance plot shows which features caused the largest validation ROC-AUC drop when shuffled. The strongest feature was M-stage_M0 with a mean ROC-AUC drop of 0.085. Other important features included M-stage_Unknown, Alcohol Consumption, Alcohol Consumption_missing, Tobacco Consumption, Performance Status, and Tobacco Consumption_missing.

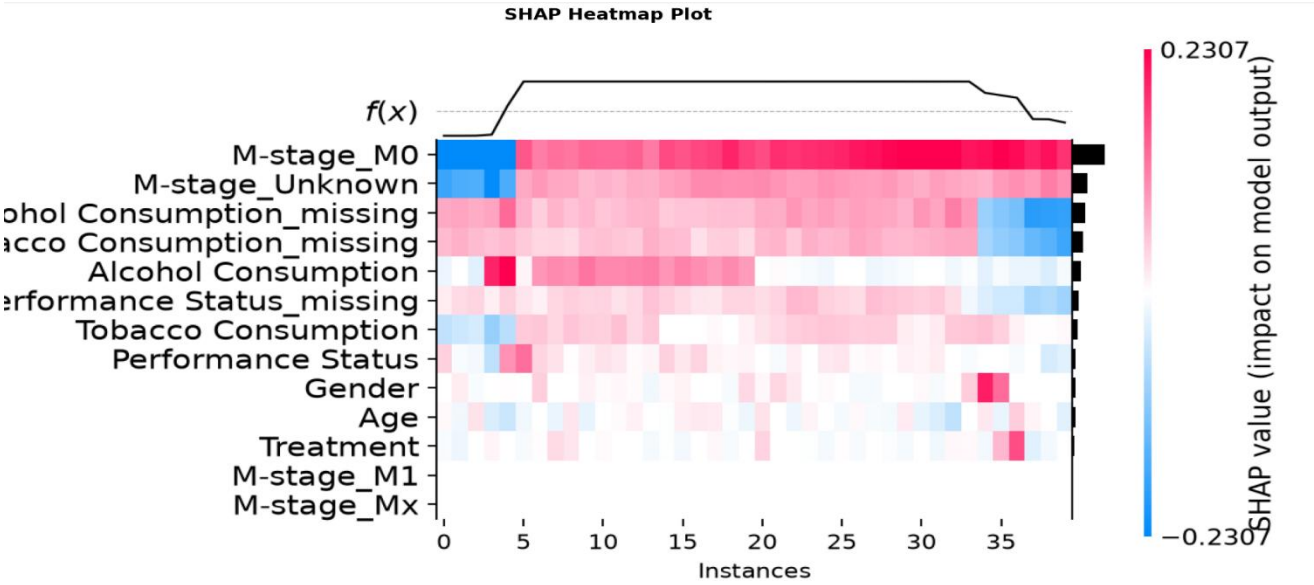
The SHAP summary plot supports the permutation importance result. The top SHAP features were M-stage_M0, M-stage_Unknown, Alcohol Consumption_missing, Tobacco Consumption_missing, Alcohol Consumption, Performance Status_missing, Tobacco Consumption, Performance Status, Gender, and Age.





The automatically selected dependence feature was M-stage_M0. This plot shows how the SHAP contribution changes when M-stage_M0 changes. Because M-stage_M0 is one-hot encoded, the plot mainly compares patients with M-stage_M0 = 0 against M-stage_M0 = 1. The feature has a clear contribution to the predicted HPV probability.

The training loss decreased from 0.659 at the first epoch to 0.178 at the final epoch. Validation loss decreased from 0.599 to 0.135. Validation ROC-AUC finished around 0.978 in the training history. This suggests the clinical model learned a stable signal and did not show severe overfitting in the saved training curve.



The SHAP heatmap shows patient-level explanations across the validation samples. M-stage_M0 and M-stage_Unknown are dominant rows, meaning they repeatedly influence predictions across many patients. Alcohol/tobacco missingness and consumption variables also contribute across multiple samples.

PET ONLY :

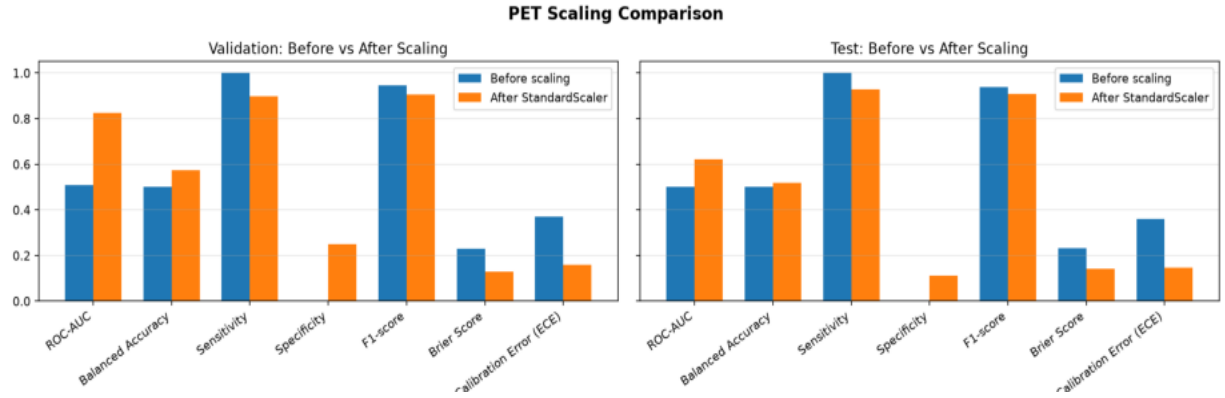
Even after applying SMOTE and SMOTENC to address class imbalance in the PET-only experiment, the model continued to exhibit poor predictive performance. Since these resampling techniques did not produce a meaningful improvement, we decided to apply standard feature scaling using the StandardScaler, fitting it only on the training set and subsequently transforming the validation and test sets using the same fitted scaler. This additional preprocessing step was performed to evaluate whether feature standardization could improve the model's learning process and lead to better predictive performance.

StandardScaler standardizes each numerical feature by subtracting its mean and dividing by its standard deviation, resulting in features with a mean of 0 and a standard deviation of 1. The scaler is fitted only on the training set, and the same transformation is then applied to the validation and test sets to prevent data leakage. This ensures that all input features are on a comparable scale, which facilitates more stable and efficient training of the Multilayer Perceptron (MLP) model.

Before vs After Scaling Metrics

Experiment	Split	ROC-AUC	Bal Acc	Sens	Spec	F1	Brier	ECE
Before scaling	Validation	0.507	0.500	1.000	0.000	0.945	0.230	0.370
Before scaling	Test	0.500	0.500	1.000	0.000	0.939	0.232	0.361
After StandardSca	Validation	0.826	0.574	0.899	0.250	0.905	0.130	0.159
After StandardSca	Test	0.621	0.519	0.928	0.111	0.908	0.141	0.147

Before vs After Scaling Plot



Before scaling, the PET-only model looked accurate only because the dataset contains many HPV-positive patients. Validation accuracy was 0.896 and test accuracy was 0.885, but Balanced Accuracy was only 0.500 on both splits. This means the model was not truly separating the two classes.

Accuracy before scaling is misleading. The model had validation accuracy of 0.896 and test accuracy of 0.885, but this happened because most patients are HPV positive and the model predicted every patient as HPV positive. So the high accuracy did not mean the model was clinically balanced.

Balanced Accuracy before scaling showed the real problem. It was 0.500 on validation and 0.500 on test, which is chance-level performance for balanced class evaluation. This confirmed that the model ignored the HPV-negative class.

The main problem before scaling was specificity. Specificity was 0.000 on validation and 0.000 on test. The confusion matrices were [[0, 8], [0, 69]] on validation and [[0, 9], [0, 69]] on test, meaning the model predicted every patient as HPV positive.

After StandardScaler, the PET model improved. Validation ROC-AUC increased from 0.507 to 0.826, and test ROC-AUC increased from 0.500 to 0.621. This means the scaled PET features helped the MLP rank HPV-positive and HPV-negative patients better.

After scaling, accuracy decreased slightly because the model stopped predicting everyone as positive. Validation accuracy changed from 0.896 to 0.831, and test accuracy changed from 0.885 to 0.833. This lower accuracy is not automatically bad, because the model started identifying some HPV-negative patients.

Balanced Accuracy improved after scaling. Validation Balanced Accuracy improved from 0.500 to 0.574, and test Balanced Accuracy improved from 0.500 to 0.519. The improvement is modest, but it shows that scaling helped the model begin to handle both classes instead of only the majority HPV-positive class.

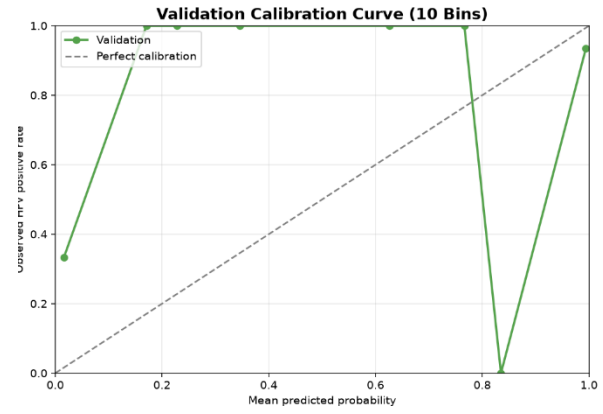
Specificity also improved after scaling, but it remained weak. Validation specificity improved from 0.000 to 0.250, and test specificity improved from 0.000 to 0.111. The test model correctly identified only 1 of 9 HPV-negative patients after scaling, so PET-only is still not strong for detecting HPV-negative cases.

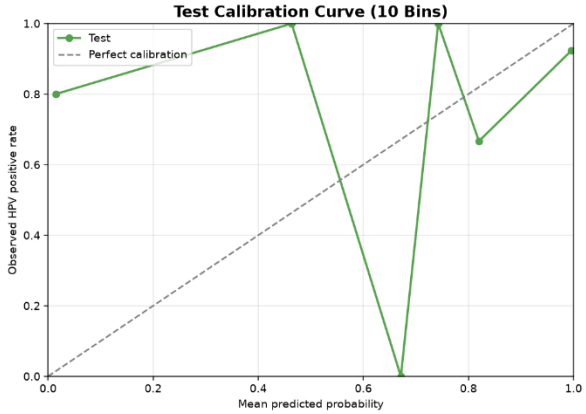
Validation Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	3	0.016	0.333	0.317
[0.1, 0.2)	2	0.171	1.000	0.829
[0.2, 0.3)	2	0.228	1.000	0.772
[0.3, 0.4)	2	0.346	1.000	0.654
[0.4, 0.5)	0			
[0.5, 0.6)	0			
[0.6, 0.7)	2	0.627	1.000	0.373
[0.7, 0.8)	3	0.766	1.000	0.234
[0.8, 0.9)	2	0.835	0.000	0.835
[0.9, 1.0]	61	0.994	0.934	0.060

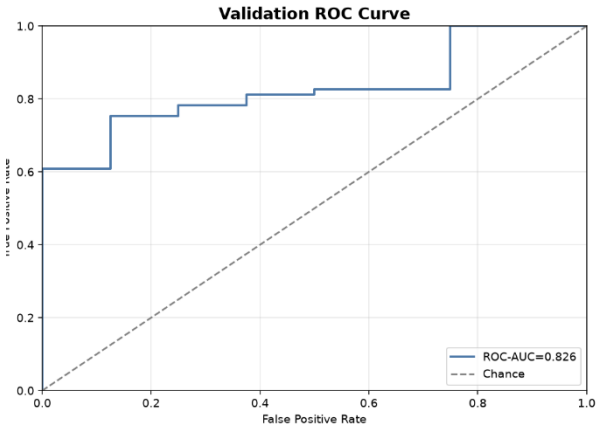
Test Calibration Table (10 Bins)

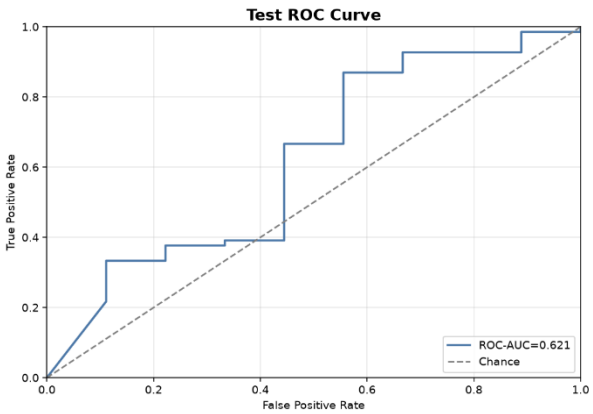
Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	5	0.014	0.800	0.786
[0.1, 0.2)	0			
[0.2, 0.3)	0			
[0.3, 0.4)	0			
[0.4, 0.5)	1	0.464	1.000	0.536
[0.5, 0.6)	0			
[0.6, 0.7)	2	0.671	0.000	0.671
[0.7, 0.8)	2	0.742	1.000	0.258
[0.8, 0.9)	3	0.820	0.667	0.153
[0.9, 1.0]	65	0.996	0.923	0.073



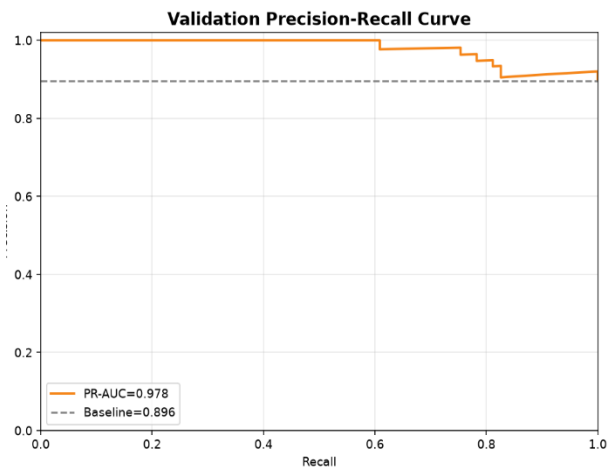


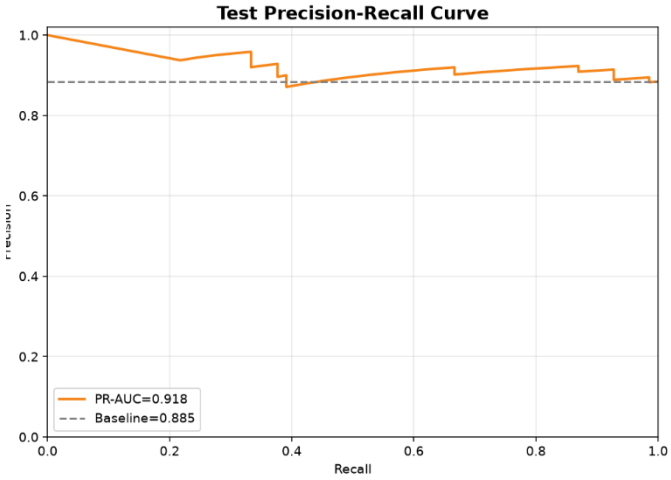
Scaling improved calibration clearly. Validation ECE improved from 0.370 to 0.159, and test ECE improved from 0.361 to 0.147. Brier Score improved from 0.230 to 0.130 on validation and from 0.232 to 0.141 on test. The probabilities are still not perfect, but the scaled PET model is much better calibrated than the unscaled PET model.



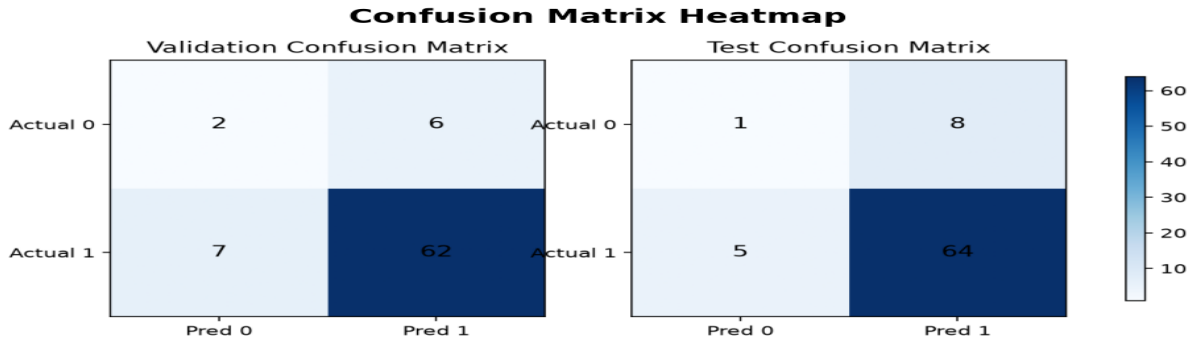


After scaling, the validation ROC-AUC is 0.826 and the test ROC-AUC is 0.621. This means scaling helped the model rank HPV-positive and HPV-negative patients better. However, the gap between validation and test is large. The PET model has weaker generalization than the clinical model.



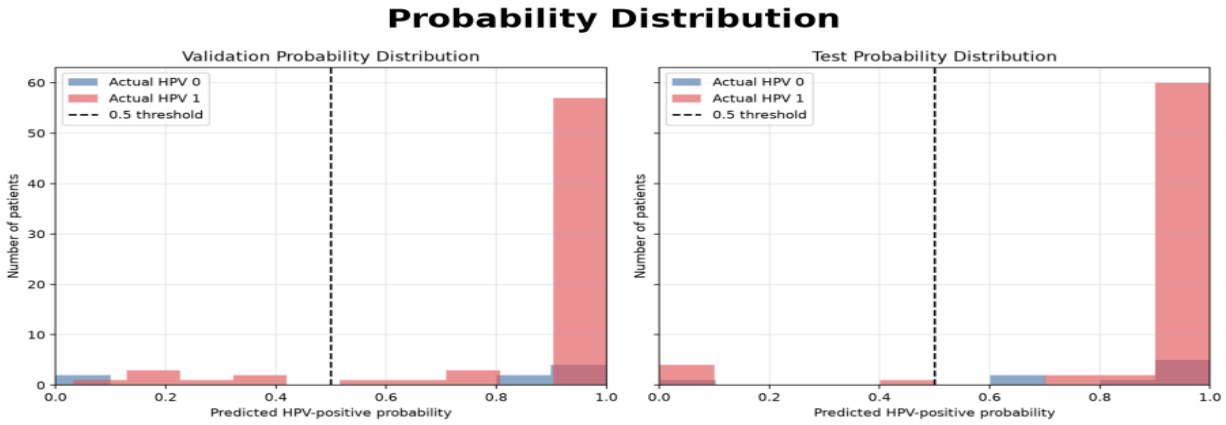


The PR-AUC is 0.978 on validation and 0.918 on test after scaling. These values are high because HPV-positive patients are the majority class and the model is still biased toward predicting positives. The PR curve is useful, but it should be interpreted together with specificity and Balanced Accuracy.

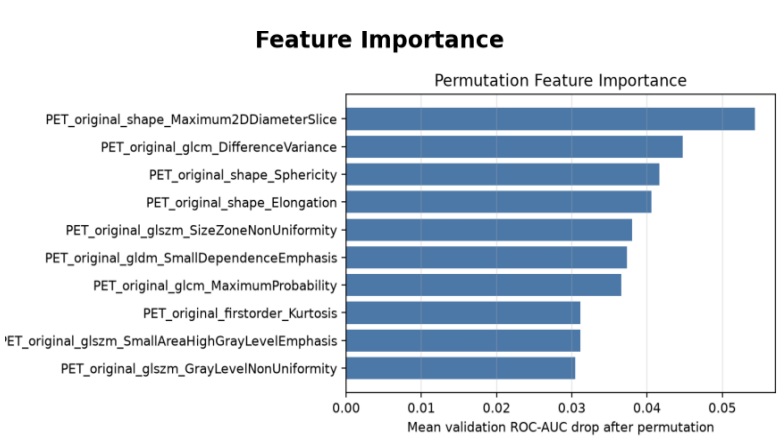


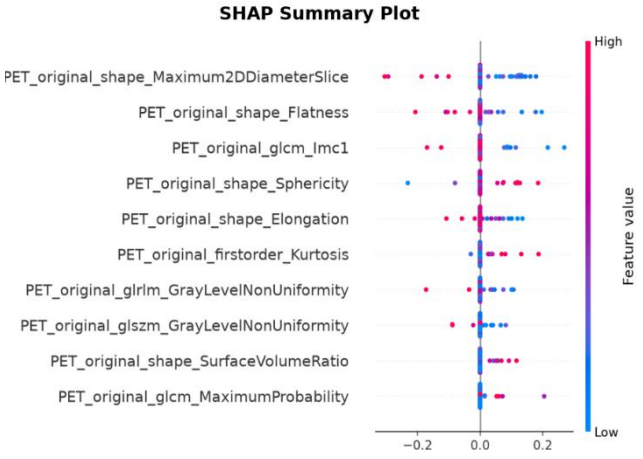
Before scaling, the PET model had confusion matrices [[0, 8], [0, 69]] on validation and [[0, 9], [0, 69]] on test. It detected every HPV-positive patient but detected no HPV-negative patient.

After scaling, validation changed to [[2, 6], [7, 62]], and test changed to [[1, 8], [5, 64]]. This is an improvement because the model now identifies some HPV-negative patients, but it still misses most HPV-negative patients.

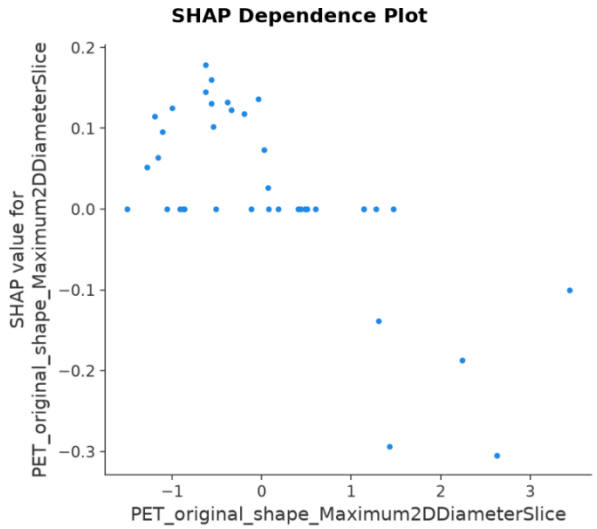


The probability distribution plot still shows many patients receiving high HPV-positive probabilities. This explains why sensitivity remains high but specificity remains low. Scaling improves separation, but the positive and negative distributions still overlap strongly.





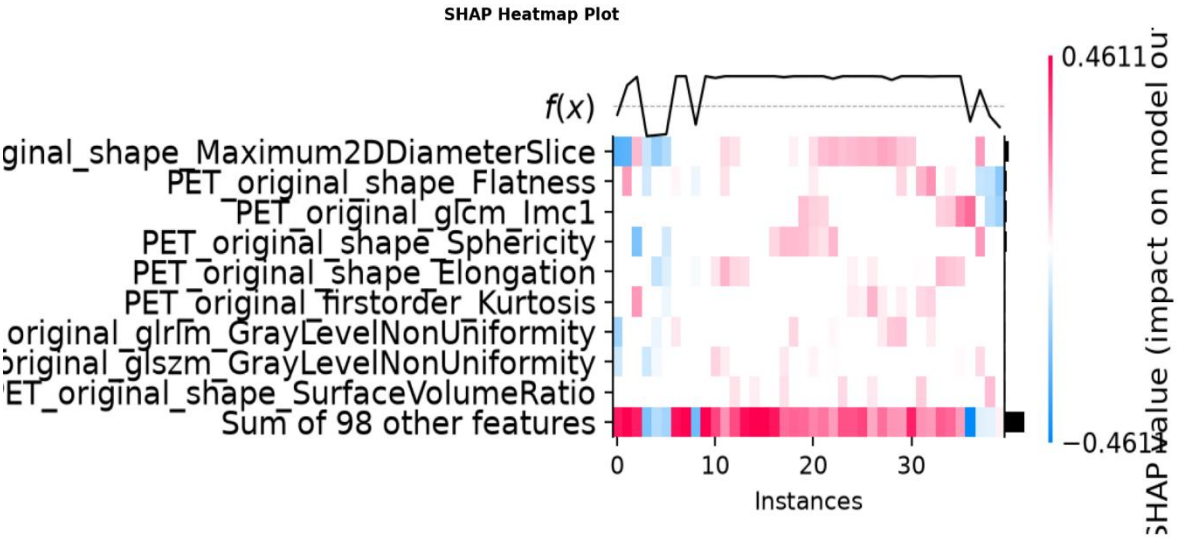
The strongest SHAP feature was PET_original_shape_Maximum2DDiameterSlice. Other important SHAP features included PET_original_shape_Flatness, PET_original_glcm_Imc1, PET_original_shape_Sphericity, PET_original_shape_Elongation, PET_original_firstorder_Kurtosis, PET_original_glrlm_GrayLevelNonUniformity, PET_original_glszm_GrayLevelNonUniformity, PET_original_shape_SurfaceVolumeRatio, and PET_original_glcm_MaximumProbability.





The automatically selected dependence feature was PET_original_shape_Maximum2DDiameterSlice. This plot shows that this tumor shape/size-related feature has a visible effect on the model output. The relationship is not perfectly linear, which is expected for an MLP.

The training loss decreased from 0.665 to 0.073, while validation loss ended around 0.463. Training ROC-AUC became very high, but validation ROC-AUC was lower. This indicates overfitting: the PET model learns the training data strongly, but the validation and test results are weaker.

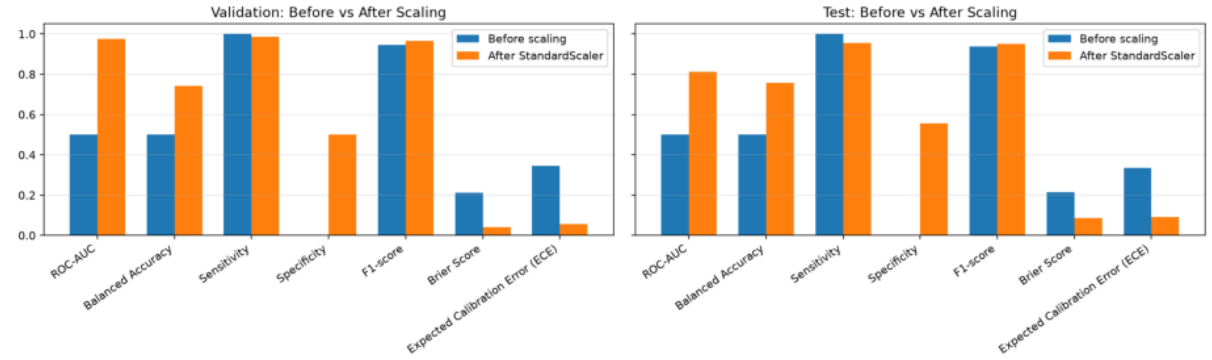


The SHAP heatmap shows that PET_original_shape_Maximum2DDiameterSlice repeatedly influences many validation patients. Other PET shape and texture features also contribute, but the heatmap suggests the PET model is less stable and less clinically clean than the clinical-only model.

MERGED ONLY :

Before vs After Scaling Metrics

Experiment	Split	ROC-AUC	Bal Acc	Sens	Spec	F1	Brier	ECE
Before scaling	Validation	0.500	0.500	1.000	0.000	0.945	0.213	0.346
Before scaling	Test	0.500	0.500	1.000	0.000	0.939	0.214	0.335
fter StandardSca	Validation	0.976	0.743	0.986	0.500	0.965	0.041	0.056
fter StandardSca	Test	0.813	0.756	0.957	0.556	0.950	0.087	0.092



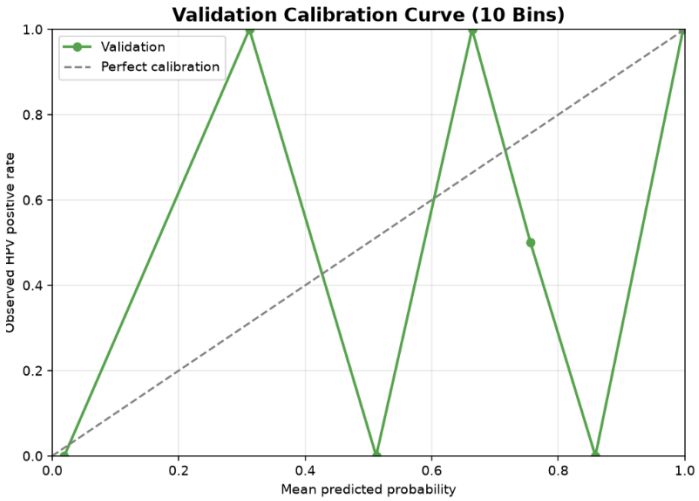
- **Before PET scaling:**
  - High accuracy (0.896 validation, 0.885 test) was misleading because the model predicted every patient as HPV-positive.
  - Balanced Accuracy was only **0.500** on both validation and test.
  - Specificity was **0.000**, meaning no HPV-negative patients were correctly identified.
- **After PET-column scaling:**
  - ROC-AUC improved from **0.500 → 0.976** (validation) and **0.500 → 0.813** (test).
  - Accuracy increased to **0.935** (validation) and **0.910** (test).
  - Balanced Accuracy improved to **0.743** (validation) and **0.756** (test).
  - High sensitivity was maintained (**0.986** validation, **0.957** test).
  - Specificity increased to **0.500** (validation) and **0.556** (test).
  - Calibration improved, with the test **Brier Score** decreasing from **0.214 to 0.087** and **ECE** from **0.335 to 0.092**.
- **Overall:** Scaling only the PET features resolved the scale issue, allowing the merged model to effectively use both clinical and PET information, making it balanced, well-calibrated, and competitive with the clinical-only model.

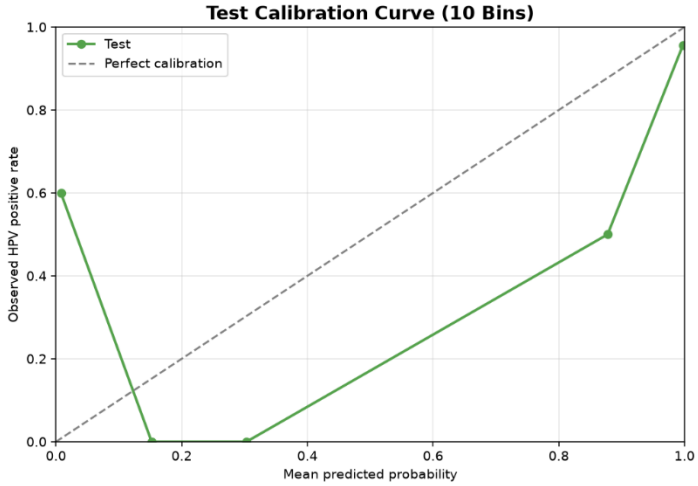
Validation Calibration Table (10 Bins)

Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	4	0.019	0.000	0.019
[0.1, 0.2)	0			
[0.2, 0.3)	0			
[0.3, 0.4)	1	0.312	1.000	0.688
[0.4, 0.5)	0			
[0.5, 0.6)	1	0.512	0.000	0.512
[0.6, 0.7)	2	0.664	1.000	0.336
[0.7, 0.8)	2	0.756	0.500	0.256
[0.8, 0.9)	2	0.859	0.000	0.859
[0.9, 1.0]	65	0.997	1.000	0.003

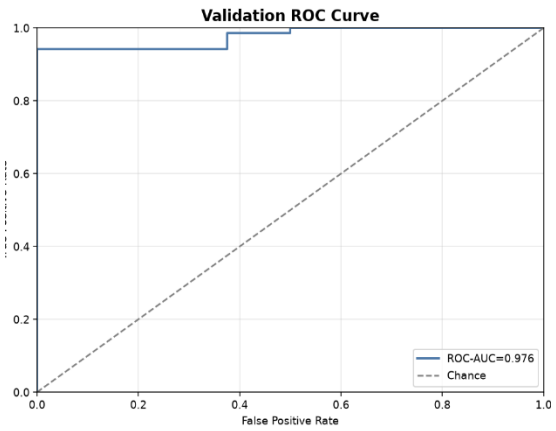
Test Calibration Table (10 Bins)

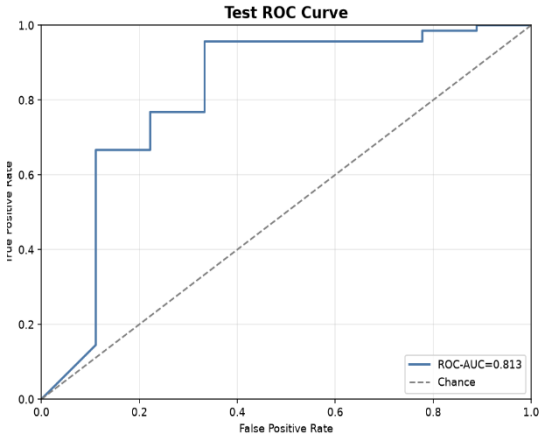
Bin	N	Mean pred	Observed +	Gap
[0.0, 0.1)	5	0.008	0.600	0.592
[0.1, 0.2)	2	0.152	0.000	0.152
[0.2, 0.3)	0			
[0.3, 0.4)	1	0.303	0.000	0.303
[0.4, 0.5)	0			
[0.5, 0.6)	0			
[0.6, 0.7)	0			
[0.7, 0.8)	0			
[0.8, 0.9)	2	0.878	0.500	0.378
[0.9, 1.0]	68	0.997	0.956	0.042



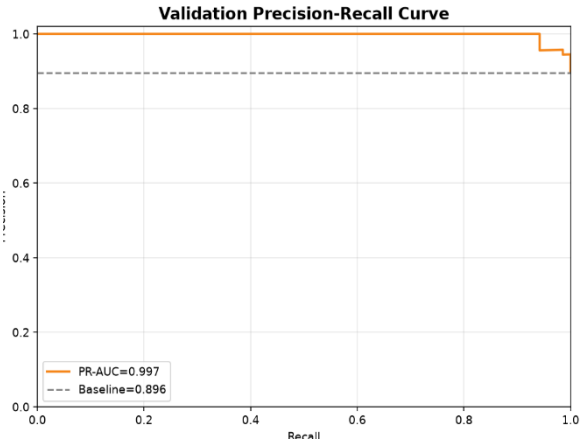


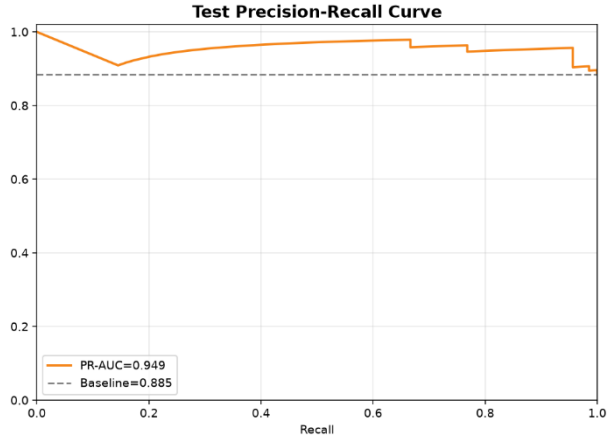
Calibration improved strongly after PET-column scaling. Validation ECE improved from 0.346 to 0.056, and test ECE improved from 0.335 to 0.092. Brier Score improved from 0.213 to 0.041 on validation and from 0.214 to 0.087 on test. This means the scaled merged model gives much more reliable probabilities than the unscaled merged model.



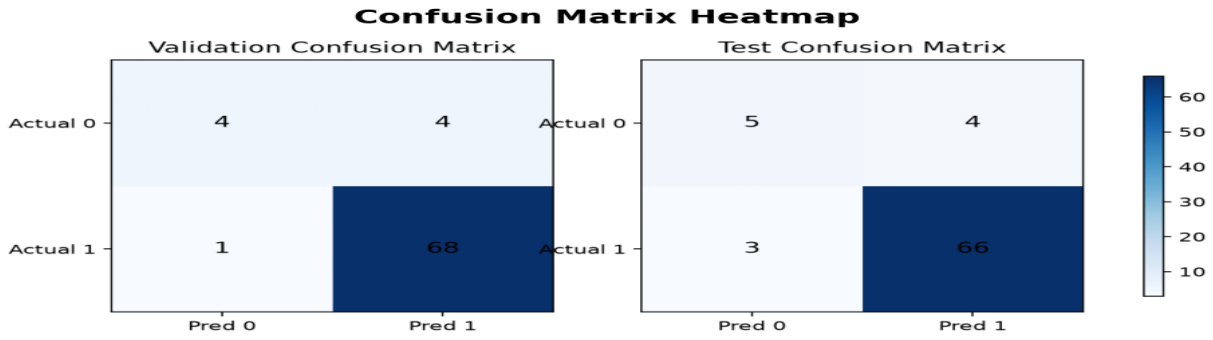


After PET-column scaling, validation ROC-AUC is 0.976 and test ROC-AUC is 0.813. The validation curve is very strong and the test curve is still clearly better than chance. The model now ranks HPV-positive patie





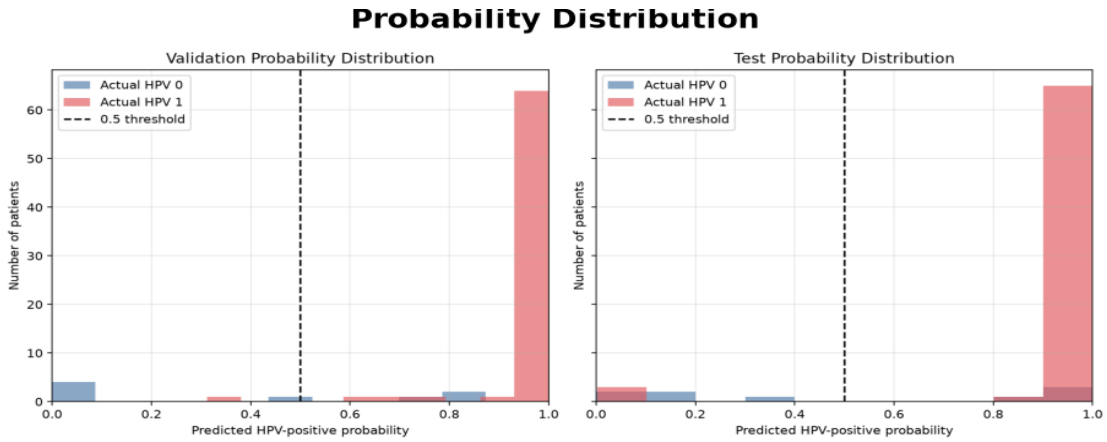
After scaling, PR-AUC is 0.997 on validation and 0.949 on test. This shows strong positive-class performance. Because the dataset is imbalanced toward HPV-positive patients, PR-AUC should be interpreted together with specificity. The merged model has better specificity than PET-only, so the strong PR-AUC is more meaningful here.nts above HPV-negative patients much better than before scaling.



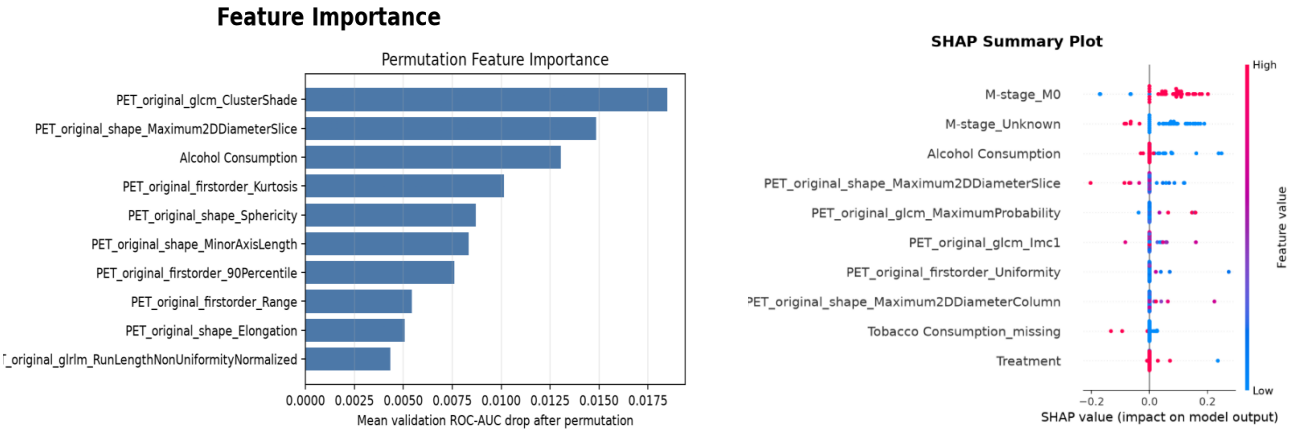
Before scaling, the merged model detected no HPV-negative patients: validation was [[0, 8], [0, 69]] and test was [[0, 9], [0, 69]].

After scaling, validation improved to [[4, 4], [1, 68]], meaning it correctly detected 4 of 8 HPV-negative patients and 68 of 69 HPV-positive patients. Test improved to [[5, 4], [3, 66]], meaning it correctly detected 5 of 9 HPV-negative patients and 66 of 69 HPV-positive patients.

This is a strong practical improvement. The model keeps high sensitivity while gaining real specificity.

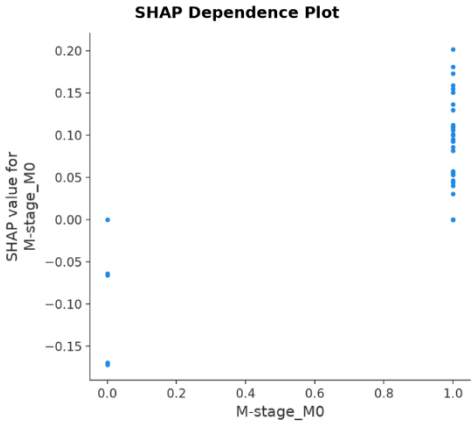


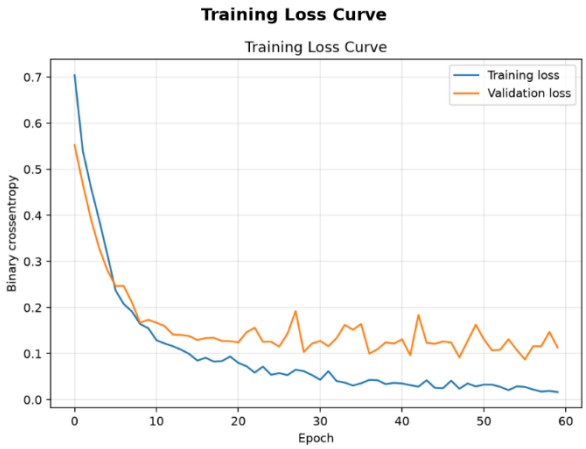
The probability distribution plot after scaling shows better separation than the unscaled merged model. Many HPV-positive patients still receive high predicted probabilities, but more HPV-negative patients now receive lower probabilities. This matches the improvement in specificity from 0.000 to 0.500 on validation and from 0.000 to 0.556 on test.



This is the best sign from the merged model: both clinical and PET features contribute. The model is not relying only on radiomics or only on clinical variables.

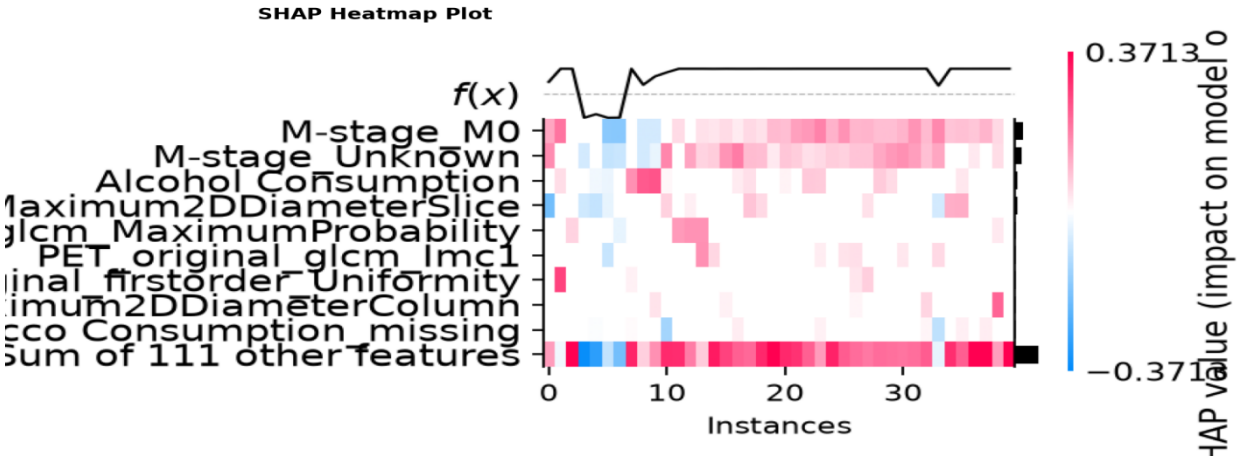
ol





The automatically selected dependence feature was M-stage_M0. This means that, even in the merged model, clinical tumor-stage information remains one of the most influential drivers of the MLP output. The PET features improve the model, but M-stage still carries major explanatory power.

The training loss decreased from 0.704 to 0.017, while validation loss ended around 0.113. Training ROC-AUC became almost perfect, and validation ROC-AUC ended around 0.976 in the training history. This suggests the model learned a strong signal. The large gap between training loss and validation loss suggests some overfitting risk, but the test ROC-AUC of 0.813 and test Balanced Accuracy of 0.756 show that the model still generalizes much better than the unscaled merged model.



he SHAP heatmap shows patient-level contributions from both clinical and PET variables. M-stage_M0 and M-stage_Unknown remain important across many validation samples. PET shape and texture features also appear in the heatmap, showing that the scaled PET features add useful information instead of overwhelming the clinical signal.